

Natural Language Processing (NLP)

Text Processing

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**National
Supercomputin
g Mission**



**Centre for
Development of
Advanced Computing**

Text Processing (Extract & Cleaning)

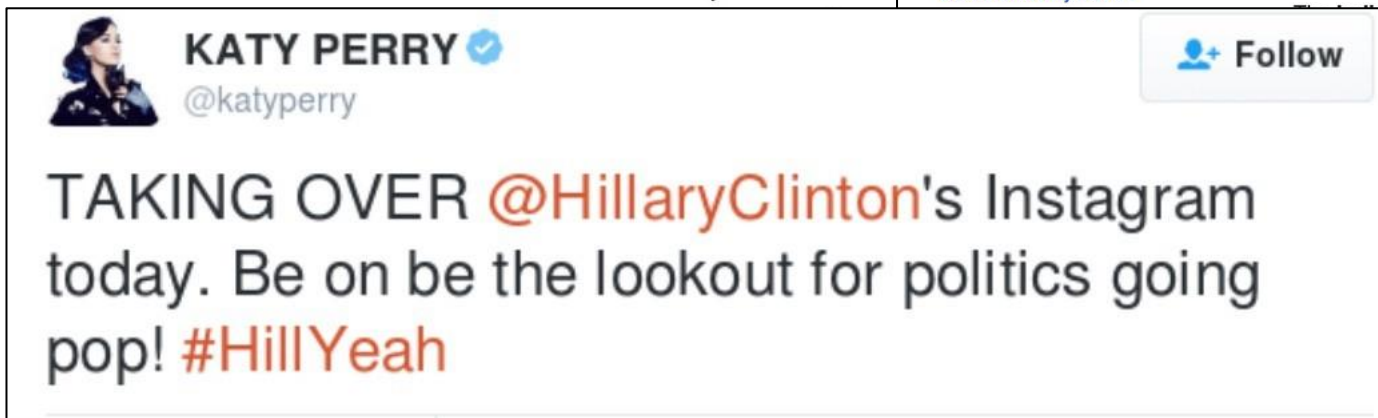
- **How to convert natural language documents into an easy-for-computer format?**
 - If data is not in **linear plain-text** format (e.g., ASCII, UTF-8), it needs to be converted (e.g., from PDF, Word, HTML)
- Words can be simply misspelled or in various forms
 - plural/singular (e.g., car, cars, foot, feet, mouse, mice)
 - tense (e.g., go, went, say, said)
 - adjective/adverb (e.g., active, actively, rapid, rapidly)
 - ...
- Issues and solutions are often highly language-specific (e.g., diacritics in German, accents in French)

Corpora

- Computer readable collection of text or speech
- Different dimensions of corpora
 - Language
 - Genre
 - Time

Inductive
Attenuation

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Space Research Organisation, commonly referred to as **ISRO**^[a] is the national **space agency** of India. It operates as the primary and development arm of the **Department of Space** (DoS), which is overseen by the **Prime Minister of India**, while the **Chairman of ISRO** is the executive of DoS.

ISRO is primarily responsible for performing tasks related to space-based **space exploration**, international space cooperation and the development of related technologies.^[1] It is one of the six government space agencies in the world which possess full launch capabilities, deploy **cryogenic** rockets to launch **extraterrestrial missions** and operate a large fleet of **artificial satellites**.^{[3][b]}

ISRO was previously the **Indian National Committee for Space Research** (INCOSPAR), set up under **Jawaharlal Nehru** in 1962 recognising the need for



tract. The sequence homology of the Sendai virus (SeV) and the Sendai virus (SeV) have made Sendai virus (SeV) immunization with Sendai virus (SeV) promotes an immune response in non-human primates that is protective against HPIV-1 [13,14] and clinical trials are underway to determine the efficacy of this

> Applications

> International cooperation

R

(-) vRNA
GP42-H1

rule of six) generating the recombinant Sendai GP42-GFP [19] or GP42-H1 vectors (iii). B) Recombinant GP42-H1 virus was cultured in BSR-T7 cells for three passages. Cell free supernatant containing virus

Tokenization

- Tokenization **splits a text into tokens**

'Who in the world am I?' Ah, that's the great puzzle!

Who	in	the	world	am	I	Ah	thats	the	great	puzzle
-----	----	-----	-------	----	---	----	-------	-----	-------	--------

- A **type** is a class of all tokens with the same character sequence
- A **term** is a (possibly normalized) type that is included into an NLP system's dictionary
- **Basic tokenization method**
 - Remove punctuation (e.g., commas, fullstops)
 - Split at white spaces (e.g., spaces, tabulators, newlines)

Issues with Tokenization

- **Language- and content-dependent**

- Boys' => Boys vs. can't => can t
- iitpkd.ac.in/moodle and support@ebay.com
- co-ordinates vs. good-looking man
- straight forward, white space, Los Angeles
- l'ensemble and un ensemble
- Compounds: Lebensversicherungsgesellschaftsangestellter
- No spaces at all (e.g., major East Asian languages)

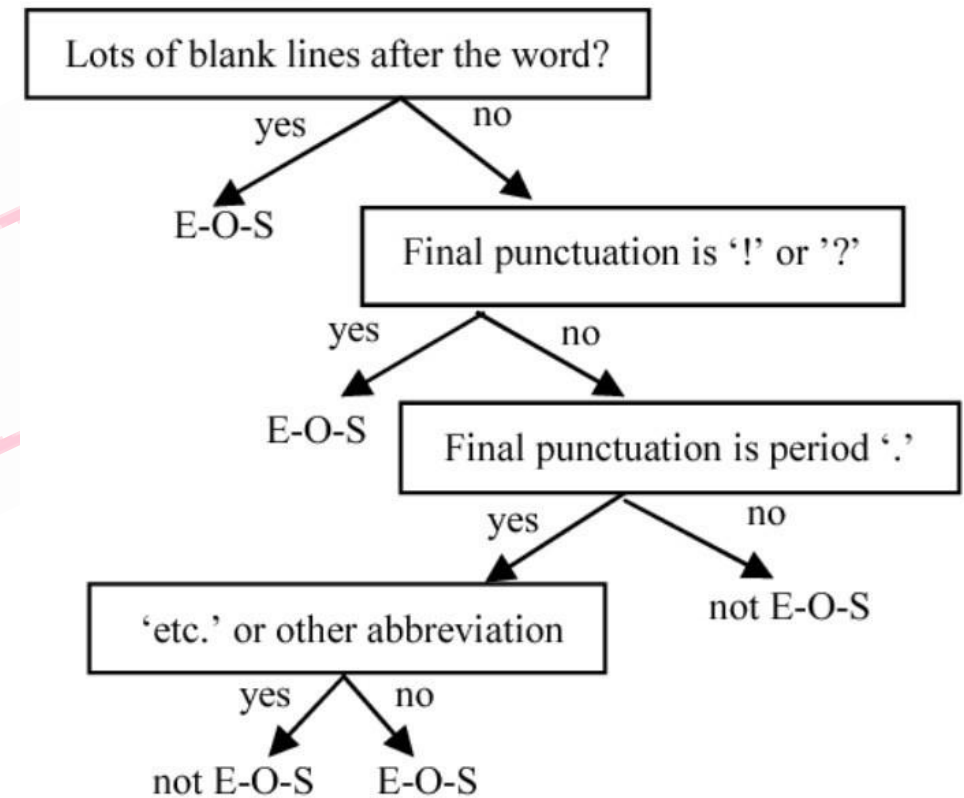
Sentence Segmentation

Theodor Seuss Geisel was an American children's author and cartoonist. He is known for his work writing and illustrating more than 60 books under the pen name Dr. Seuss. He completed his Ph.D. in Lincoln College, Oxford in English literature.

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Stop Words

- Stop words are **very frequent words that carry no information** and are thus excluded from the system's dictionary (e.g., **a, the, and, are, as, be, by, for, from**)
- Can be defined explicitly (e.g., with a list) or implicitly (e.g., as the k most frequent terms in the collection)
- Mostly the function words in large corpus
- Removing them saves significant space
 - **can cause problems**
 - to be or not to be, the who, etc.
 - “president of the united states”, “with or without you”, etc.
- Current trend towards shorter or no-stop-word lists

Stemming

- **Variations of words could be grouped together**
e.g., plurals, adverbial forms, verb tenses
- A crude heuristic is to cut the ends of words
(e.g., ponies => poni, individual => individu)
- Word stem is not necessarily a proper word
- **Variations of the same word ideally map to same unique stem**
- Popular stemming algorithms for English
 - Porter (<http://tartarus.org/martin/PorterStemmer/>)

Stemming

*Two households, both alike in dignity,
In fair Verona, where we lay our scene,
From ancient grudge break to new mutiny,
Where civil blood makes civil hands unclean.
From forth the fatal loins of these two foes*



*Two household, both alik in digniti,
In fair Verona, where we lay our scene,
From ancient grudg break to new mutini,
Where civil blood make civil hand unclean.
From forth the fatal loin of these two foe*

Lemmatization

- Lemmatizer conducts **full morphological analysis** of the word to identify the lemma (i.e., dictionary form) of the word
- **Saw → see(Verb)/saw(Noun)**
- Word consists of **stems** and **affixes**
 - **Unhappy, Disrespectful, runners-up**
- **Morphemes – smallest meaningful units** (**Free** or **Bound**)
 - Unhappy - 2 morphemes un (not), happy
 - Disrespectful- dis(not/lack), respect, ful(having the characteristic of)

Other Processing Ideas

- **Diacritics** (e.g., ü, ø, à, ě)
 - Remove/normalize diacritics: ü => u, å => a, ø => o
 - Diacritics are sometimes typed using multiple characters: für => fuer
- **Lower/upper-casing**
 - Discard case information (e.g., United States => united states)
- **n-grams as sequences of n characters** (inter- or intra-word) are useful for Asian (CJK) languages without clear word spaces

Spelling Correction and Edit Distances

- users don't know how to spell, typing errors, error due to noisy transmission, etc.
- When user types in unknown, potentially misspelled word, we can try to **map it to “closest” term in dictionary**
- **Need notion of distance between terms**
 - adding extra character (e.g., hooose vs. house)
 - omitting character (e.g., huse)
 - using wrong character (e.g., hiuse)
 - as-heard spelling (e.g., kwia vs. choir)

Levenshtein Edit Distance

- **Levenshtein edit distance** between two strings x and y is the minimal number of **edit operations** (**insert**, **replace**, **delete**) required to transform x into y
- The **minimal number of operations** $m[i, j]$ to transform the prefix substring $x[1:i]$ into $y[1:j]$ is defined via

$$m[i, j] = \min \begin{cases} m[i-1, j-1] + (x[i] = y[j] ? 0 : 1) & \text{(replace } x[i]?) \\ m[i-1, j] + 1 & \text{(delete } x[i]) \\ m[i, j-1] + 1 & \text{(insert } y[j]) \end{cases}$$

- **Examples:**
 - $d(\text{hooose}, \text{house}) = 1$
 - $d(\text{house}, \text{rose}) = 2$

Levenshtein Edit Distance: Dynamic Programming

- Example:** cat vs. kate

		j →				
		-	k	a	t	e
i ↓	-	0	1	2	3	4
	c	1	?			
	a	2				
	t	3				

Now looking at (1,1), i.e., distance of string "c" with "k"

What choices do we have?

- **replace:** replacing "c" with "k" and add cost of 1 to cost of $m[0,0]$, i.e., distance of empty strings 1
- **delete:** delete "c" and add 1 to cost of $m[0,1]$, i.e., distance of empty string to "k" 2
- **insert:** Insert "k" and add 1 to cost of $m[1,0]$, i.e., distance of string "c" to empty string 2

$$m[i, j] = \min \begin{cases} m[i-1, j-1] + (x[i] = y[j] ? 0 : 1) & \text{(replace } x[i]?) \\ m[i-1, j] + 1 & \text{(delete } x[i]) \\ m[i, j-1] + 1 & \text{(insert } y[j]) \end{cases}$$



Levenshtein Edit Distance: Dynamic Programming

- Example:** cat vs. kate

		j →				
		-	k	a	t	e
i ↓	-	0	1	2	3	4
	c	1	1	?		
	a	2				
	t	3				

Now looking at (1,2), i.e., distance of string "c" to "ka"

What choices do we have?

- replace:** replacing "c" with "a" => add cost of 1 to cost of $m[0,1]$, i.e., distance of empty string with "k" 2
- delete:** delete "c" => add 1 to cost of $m[0,2]$, i.e., distance of empty string to "ka"? 3
- insert:** insert "a" => add 1 to cost of $m[1,1]$, i.e., distance of string "c" to "k" 2

$$m[i, j] = \min \begin{cases} m[i-1, j-1] + (x[i] = y[j] ? 0 : 1) & \text{(replace } x[i]?) \\ m[i-1, j] + 1 & \text{(delete } x[i]) \\ m[i, j-1] + 1 & \text{(insert } y[j]) \end{cases}$$



Levenshtein Edit Distance: Dynamic Programming

- **Example:** cat vs. kate

		 j				
		-	k	a	t	e
 i	-	0	1	2	3	4
	c	1	1	2	3	4
	a	2	2	1	2	3
	t	3	3	2	1	2

feture -> feature
feture -> future

} Similar MED

$$m[i, j] = \min \begin{cases} m[i-1, j-1] + (x[i] = y[j] ? 0 : 1) & \text{(replace } x[i]?) \\ m[i-1, j] + 1 & \text{(delete } x[i]) \\ m[i, j-1] + 1 & \text{(insert } y[j]) \end{cases}$$


Basic Text Pre-Processing Pipeline

- Tokenization
 - Sentence Segmentation
- Stemming/Lemmatization
- Stop word removal
- Spelling Correction

Thank You

References:

Chris Manning and Hinrich Schütze, 1999. Foundations of Statistical Natural Language Processing, MIT Press. ISBN: 9780262133609 . Online available [here](#).

Daniel Jurafsky and James Martin, 2021. Speech and Language Processing. Third Edition Draft. [Online](#)

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